Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/30/2004
PI Name:	Stebe, Kathleen J. Ph.D.		
Project Title:	Using Surfactants to Control Bubble Growth Coalescence in Nucleate Pool Boiling		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
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Zip Code:	21218-2694	**Congressional District:**	7
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Balasubramaniam, R. (NASA Glenn Research Center)		
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Task Description:	The manner in which surfactants alter the final stages of drop or bubble detachment is studied numerically using a volume of fluid code with a front tracking scheme. Drop necking and snap off in viscous fluids has universal behavior in the absence of surfactants. Necks develop in a self-similar manner regardless of initial conditions. Our results show that surfactants create conditions in drop necks that violate this universality, creating new modes of necking and detachment that depend on surfactant mass transfer dynamics. These breakup modes are also self-similar, indicating that they are also determined by local balances of viscous and capillary stresses, with the added complexity that the surface tension profiles are highly coupled to the surfactant distribution. In other cases, surfactants prevent necks from forming, causing a transition from dripping to jetting modes. To elucidate the manner in which surfactants create new break off modes with an aim to harnessing these new mechanisms in applications such as heat transfer in microgravity, drop and bubble management in microfluidics. Furthermore, these studies can be exploited in the characterization of surfactant dynamics. Buoyancy plays an important role in nucleate pool boiling on the Earth. When the heater surface is below the liquid,		

	vapor bubbles depart from the surface due to the action of buoyancy. This promotes the rate of heat transfer from the heated surface. When the heating surface is above the liquid under normal gravity, the vapor bubbles do not move away from the surface as buoyancy pushes the bubbles toward the heater. The growing bubbles soon coalesce and a regime of film boiling is achieved. In such a mode of operation the benefits of nucleate boiling are not utilized. For efficient operation of the system in the regime of nucleate boiling, the dynamics of the vapor bubbles must be understood and controlled. This is even more important in reduced gravity because in the absence of buoyancy, the primary mechanism for the removal of bubbles from the heating surface is eliminated. We propose to control the dynamics of bubbles in nucleate boiling in reduced gravity by the addition of soluble surfactants to the liquid.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Drop growth and detachment studies are currently used to infer surfactant properties such as adsorption-desorption rates or diffusivity in experiments like the drop weight technique or maximum bubble pressure techniques. The results from our studies will allow these methods to be made more robust, in that incorrect, ad hoc assumptions about drop shapes, growth modes and detachment mechansims can be eliminated. In microgravity experiments in which immiscible fluids are used to make discrete fluid elements (e.g. drop or bubble formation in microfluidics channels surrounded by other fluids) surfactants have been deminstrated in experiment to strongly alter drop formation and detachment mechanisms, without a detailed understanding of how surfactant physical chemistry dictates the observed changes. Such an understanding will be advanced by this research. In the creation of dispersions (drops, emulsions, bubbles, sprays) surfactants play a strong role in altering drop and bubble size, stability, etc,. These fields will be impacted by the results of this current study.
Task Progress:	We study the effect of surfactants on the evolution of drops from formation to break up, using a volume-of-fluid code with a front tracking scheme. This work focuses on the very last stages of drop detachment. When a viscous drop detaches in a viscous fluid, it evolves to form a distended drop that detaches via the rapid formation and pinching of a neck. Typically, two necks form, a primary neck beneath the detaching drop, and a secondary neck just above the remnant drop. In the absence of surfactants, neck pinch-off occurs at the primary neck and displays self-similar dynamics that are governed locally by the balance of capillary and viscous stresses in creeping flow. For viscous drops surrounded by viscous liquids, this regime has been predicted by scaling arguments, verified in experiment and studied using boundary integral simulations. We use these results to calibrate our code, recovering the experimental results characterizing the self-similar regime (slopes, shapes, and growth rates) over an extended viscosity ratio range. We then present results in which the addition of surfactants to the system either allows new self-similar behavior (always characterized by cones) to emerge, or altogether suppresses the formation of a neck, stabilizing the drop against detachment. In the absence of surfactants, as a drop enters the necking regime, the rate of surface dilatation is most negative just above the primary neck. When surfactants are present, surfactant accumulation occurs there. When this accumulation is sufficiently pronounced, surfactant concentrations can become comparable to maximum packing well within the primary neck, and into the elongated region. Since surface tension response is highly nonlinear at concentrations comparable to the maximum packing, necking dynamics can be significantly altered. For example, for insoluble surfactants at high enough surface concentrations, drops break at the primary neck with new self-similar shapes featuring shallower slopes above and below the neck. For soluble surfactants, even more complexity emerges. In the sorption-controlled limit with infinitely fast kinetics of surfactant exchange, drop dynamics are indistinguishable from the clean case. For slower exchange kinetics, new self-similar shapes emerge; the primary neck forms first, but thins more slowly because of surfactant accumulation, so the secondary neck thins at a similar rate, both in a self-similar manner, so primary and secondary necks are symmetric. At higher surface coverages, the secondary neck becomes the point of pinch-off, with the primary neck failing to form; the detaching drop adjoins an elongated region which merges into the pinch-point adjoining the remnant drop below. An inverted similarity solution is found, with a shallow cone above and steep cone below. Finally, regimes are identified in which the pronounced accumulation of surfactant in the elongated region prevents it from necking, and favors a transition from dripping to jetting modes.
Bibliography Type:	Description: (Last Updated: 01/16/2006)
Articles in Peer-reviewed Journals	Jin, F.; Balasubramaniam, R. ; Steb, K.J. "Surfactant Adsorption to Spherical Particles: The Intrinsic Length Scale; J. of Adhesion " none , Sep-2004